Engineering Specification: GBA Save State Interoperability Utility

Target Platform: Anbernic RG34XXSP (MuOS)
Host System: macOS (Development & Testing)

1. Hardware & OS Environment

1.1 Target Device: Anbernic RG34XXSP

Unlike its predecessors, the RG34XXSP features a distinct hardware profile that significantly benefits application development, particularly regarding memory overhead.

Component	Specification	Dev Implication
SoC	Allwinner H700 (Quad-core A53)	High performance for scripting; negligible latency.
RAM	**2GB LPDDR4**	Massive headroom. Python scripts can load entire save files into memory without risk of OOM (Out of Memory) kills.
Display	**3.4-inch, 720 x 480**	**Aspect Ratio 3:2**. This is wider than the standard 640x480 (4:3). TUI layouts must be adjusted to utilize the extra horizontal real estate (approx. 90 columns vs 80).
Input	D-Pad, Buttons	No touch. Navigation must be strictly keyboard-driven via whiptail.
Sleep	Hall Magnetic Switch	Lid close triggers sleep. File I/O must be atomic to prevent corruption if the lid is closed mid-write.

1.2 Operating System: MuOS (Linux)

MuOS on the H700 chipset is a lightweight Linux distribution. It uses a specific directory structure for user applications.

- **Python:** Python 3 is pre-installed at /usr/bin/python3.
- **UI Library:** whiptail (libnewt) is the native tool for menus and dialogs.
- **Mount Points:**
    - SD Slot 1: /mnt/mmc (System & Apps)
    - SD Slot 2: /mnt/sdcard (Roms & Saves)

2. Core Logic: The mGBA vs. VBA Conflict

The core utility function addresses the binary incompatibility between mGBA (default on MuOS) and VBA (standard on PC/Mac).

- **Standard Save (VBA/Flashcart):** Exactly 131,072 bytes (128KB) for most Pokemon games.
- **mGBA Save:** Appends a **16-byte RTC footer** containing timestamp data. Total size: 131,088 bytes.

Conversion Algorithm:

1. **Detection:** If file size is 131,088 bytes -> Identify as **mGBA**.
2. **Sanitization:** Truncate the last 16 bytes.
3. **Result:** A standard 131,072 byte file compatible with VBA.
4. *Note:* Converting VBA -> mGBA is usually not required; mGBA will accept a raw save and append a new footer upon next save. The critical path is **mGBA -> VBA**.

3. System Architecture

The application consists of a shell script entry point (for the UI) and a Python script (for the logic).

3.1 Directory Structure

MuOS requires a specific folder structure to recognize the app and its assets.

```
/mnt/mmc/MUOS/application/GBA_Fixer/
├── launch.sh # Entry point & UI Loop
├── convert.py # Binary processing logic
└── assets/
├── icon.png # 320x320px (Displayed in menu)
```

└── preview.png # 640x480px (Displayed as background)

3.2 launch.sh (The UI Controller)

This script manages the whiptail interface. It must handle the file selection dialog.

- **Resolution Handling:** With a 720px width, whiptail dialogs should be set to ~**85-90 characters wide** to look balanced.
- **File Browser:** Since whiptail does not have a built-in file picker like dialog's --fselect, we must implement a loop that lists .sav files in a menu.

4. Development & Testing Strategy (macOS + Docker)

Since you are developing on a Mac, you cannot run the MuOS shell scripts directly due to differences in sed, awk, and file paths. We will use a **Hybrid Testing Approach**.

4.1 Option A: Docker Container (Recommended)

This allows you to run the exact Linux shell environment on your Mac. There is no "Official MuOS Container," but a standard Debian slim image is 99% identical for this purpose.
Dockerfile:
Save this as Dockerfile in your project root.

Dockerfile

```
FROM python:3.11-slim-bookworm

# Install whiptail and basic utils to mimic MuOS environment
RUN apt-get update && apt-get install -y whiptail coreutils

# Create the MuOS directory structure mock
RUN mkdir -p /mnt/mmc/MUOS/application/GBA_Fixer \
   && mkdir -p /mnt/sdcard/ROMS/GBA

# Set working directory
WORKDIR /mnt/mmc/MUOS/application/GBA_Fixer

# Copy your local scripts into the container
COPY..
```

```
# Make scripts executable
RUN chmod +x launch.sh

CMD ["/bin/bash"]
```

Testing Workflow:

1. Build: docker build -t muos-test.
2. Run: docker run -it --rm -v $(pwd):/mnt/mmc/MUOS/application/GBA_Fixer muos-test /bin/bash
3. Inside Docker: Run ./launch.sh. You will see the TUI inside your Mac terminal exactly as it will appear on the device.

4.2 Option B: Local macOS Testing

If you prefer not to use Docker, you must install the dependencies via Homebrew.

1. **Install Whiptail:** brew install newt
2. **Mock Paths:** Your Python script must detect if it's running on macOS and switch paths.

Python

```
import platform
import os

IS_MACOS = platform.system() == 'Darwin'

if IS_MACOS:
    # Create a local 'mock' folder for testing
    ROOT_DIR = os.path.join(os.getcwd(), "mock_sd")
else:
    # Actual MuOS paths
    ROOT_DIR = "/mnt/sdcard"
```

5. Implementation Logic (For Claude Code)

Pass the following instructions to the coding agent to generate the implementation.

5.1 convert.py Requirements

- **Input:** Accepts a single filename as a command-line argument.
- **Validation:**
  - Check if file exists.
  - Check size.
  - If 131,088 bytes: Perform **Trim** (Remove last 16 bytes).
  - If 131,072 bytes: Return **"Already Raw"**.
  - Other sizes: Return **"Unknown Format"**.
- **Safety:**
  - Create a .bak copy before touching the original.
  - Use os.fsync() to ensure write persistence.

5.2 launch.sh Requirements

- **Header:** Include MuOS metadata comments.
  Bash
  #!/bin/sh
  # NAME: GBA Save Fixer
  # ICON: icon.png
  # CATEGORY: Utility

- **Resolution Config:** Explicitly define whiptail dimensions for 720x480.
  - Height: 20 rows
  - Width: 85 cols
- **Logic Loop:**
  1. Scan /mnt/sdcard/ROMS/GBA (and /mnt/mmc/ROMS/GBA) for .sav files.
  2. Format the list for whiptail --menu.
  3. On selection, call python3 convert.py "filename".
  4. Capture python output and display it in a whiptail --msgbox.

5.3 Deployment

1. **Zip:** Create a zip file named GBA_Fixer.zip containing the GBA_Fixer folder.
2. **Install:**
   - **Manual:** Extract to /mnt/mmc/MUOS/application/.
   - **MuOS Archive Manager:** Place zip in Archive folder on SD card and install via on-device menu.